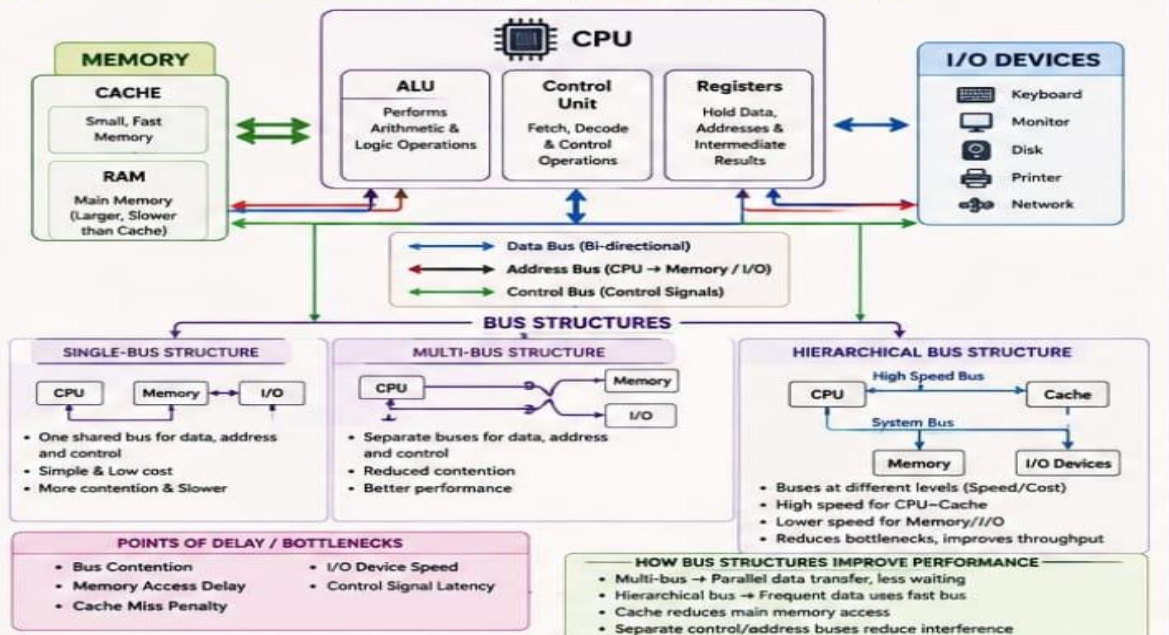


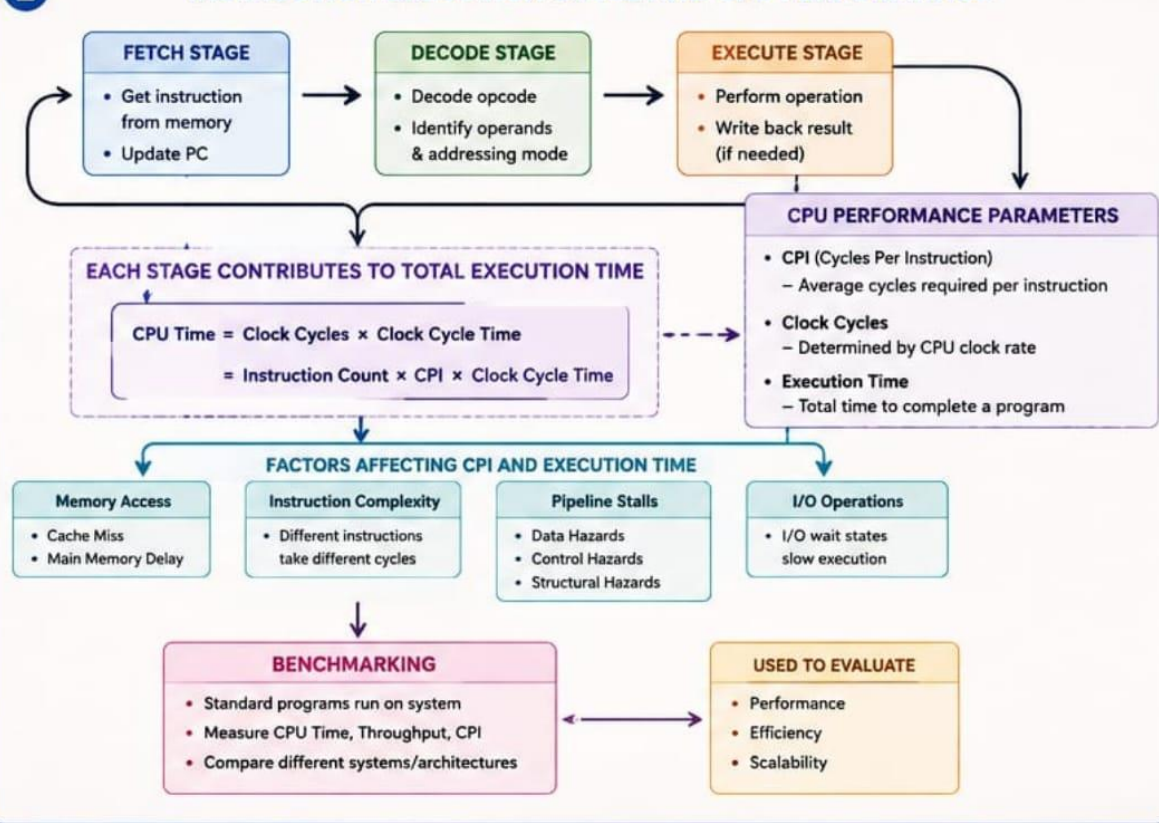
1

CPU, MEMORY, I/O AND BUS STRUCTURES



2

INSTRUCTION EXECUTION CYCLE AND CPU PERFORMANCE



3

COMPARISON OF 8085 AND 8086 ARCHITECTURES

FEATURE	8085	8086	HOW FEATURES INFLUENCE PERFORMANCE
Architecture	8-bit Microprocessor	16-bit Microprocessor	
Registers	A, B, C, D, E, H, L (8-bit) PC (16-bit), SP (16-bit) Flag Register (8-bit)	AX, BX, CX, DX (16-bit) SP, BP, SI, DI (16-bit) CS, DS, SS, ES (16-bit) IP (16-bit), Flag Register (16-bit)	 Larger Register Set (8086) → Fewer memory accesses, faster execution
Instruction Set	74 Instructions	~200+ Instructions	 More Addressing Modes (8086) → Greater flexibility, optimized code
Addressing Modes	5 (Immediate, Register, Direct, Register Indirect, Implied)	8 (Includes Indexed, Based, Based Indexed, Relative, etc.)	
Memory Addressing	16-bit Address Bus (64 KB Memory)	20-bit Address Bus (1 MB Memory)	 Segmented Memory (8086) → Supports larger programs, better multitasking
Memory Organization	Linear Memory	Segmented Memory (64 KB segments)	
Multitasking Capability	Limited	Better (due to segmentation & larger address space)	 16-bit Architecture (8086) → Higher data handling capacity & speed
Efficiency in Real-time Apps	Good for simple, dedicated control tasks	Better for complex, multi-tasking & high performance tasks	
			RESULT 8086 is more powerful, efficient and suitable for advanced and real-time applications.

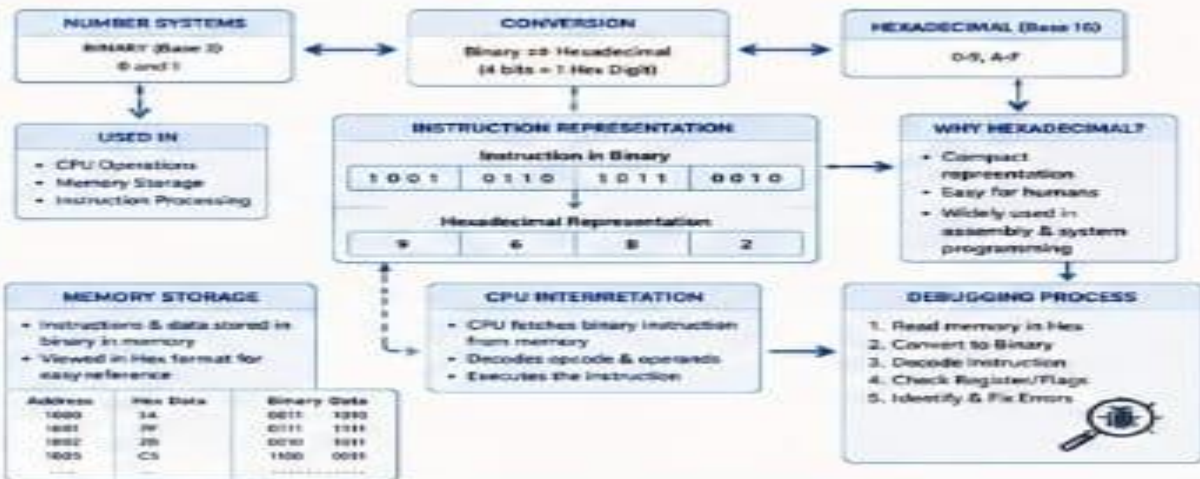
ADDRESSING MODE	DESCRIPTION	USED WITH INSTRUCTION TYPES	IMPACT	REAL-WORLD USAGE
Immediate	Operand is part of the instruction	- Data Transfer - Arithmetic	- No memory access - Larger instruction size - Fast execution	- Constant values - Initialization
Direct	Address of operand is given in instruction	- Data Transfer - Arithmetic	- One memory access - Instruction size medium	- Global variables - Fixed memory locations
Register	Operand is in CPU register	- Data Transfer - Arithmetic - Logical	- No memory access - Small instruction size - Very fast	- High speed computations - Counters
Indirect	Address of operand is in a register (or memory)	- Data Transfer - Arithmetic	- More memory access - Larger instruction size - Slower	- Pointers - Dynamic memory access
Indexed	Effective address = Base address + Index	- Data Transfer - Control instructions	- Extra computation - More memory access - Slower	- Array/Table access - Loop processing



SUMMARY

Choice of addressing mode affects memory access time, instruction size and execution speed. Proper selection improves overall program performance.

5

NUMBER SYSTEMS, INSTRUCTION REPRESENTATION
AND DEBUGGING

☆ Hexadecimal representation simplifies low-level programming, memory analysis and debugging, making it essential in system design.